



German Conference on Bioinformatics

WS8: Spatial domain identification: computational methods for discovering tissue architecture

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21.09.2025



ZMNH
Zentrum für Molekulare
Neurobiologie Hamburg



Universitätsklinikum
Hamburg-Eppendorf

Part I

Methods

Methods and datasets

GE + XY

GraphST
ConST
STAGATE
Spatial-MGCN
STMGCN

SEDR
SCGDL
SpaceFlow
CCST
spaGCN

Methods
($n = 19$)

Real datasets
($n = 30$)

Synthetic datasets
($n = 27$)

GE

Leiden
Louvain
Seurat
BayesSpace

GE + H&E

Muse
Spacell

GE + XY + H&E

stLearn
DeepST
SpaGCN+

Identifying spatial domains



Domain 1
Domain 2
Domain 3
Domain 4
Domain 5

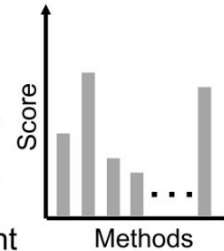
▮ Accuracy

▮ Stability

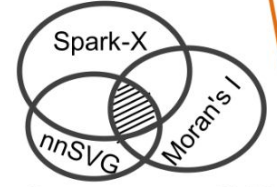
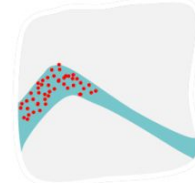
▮ Generality

▮ Scalability

▮ Refinement



Detecting domain-specific SVGs



Consensus SVGs

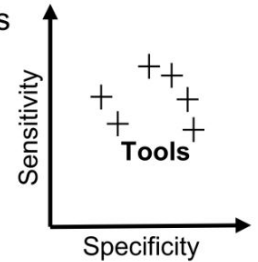
▮ No. of SVGs

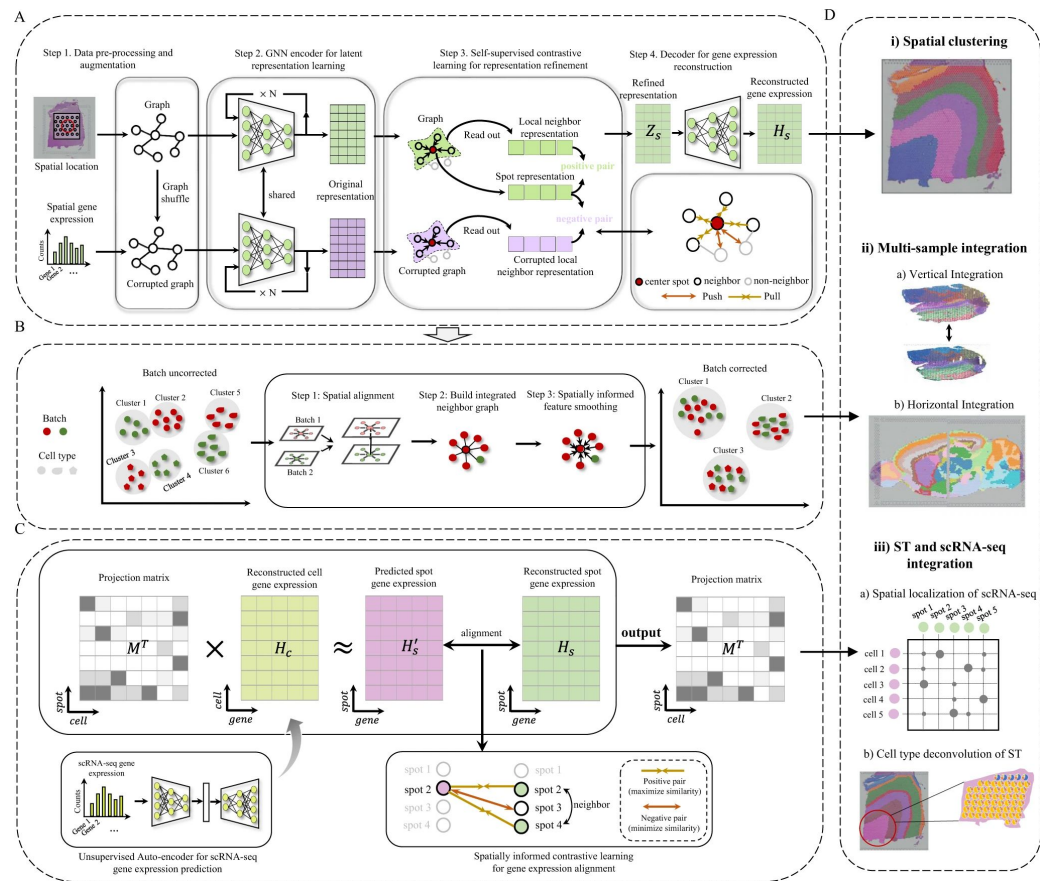
▮ Moran's I

▮ Geary's C

▮ AUPR

▮ SN / SP



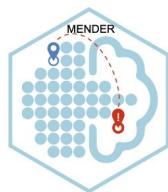
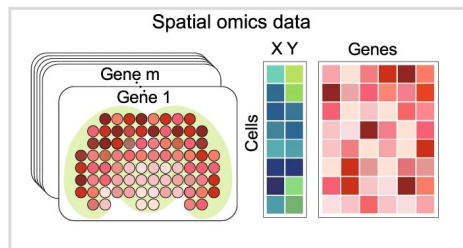


GraphST

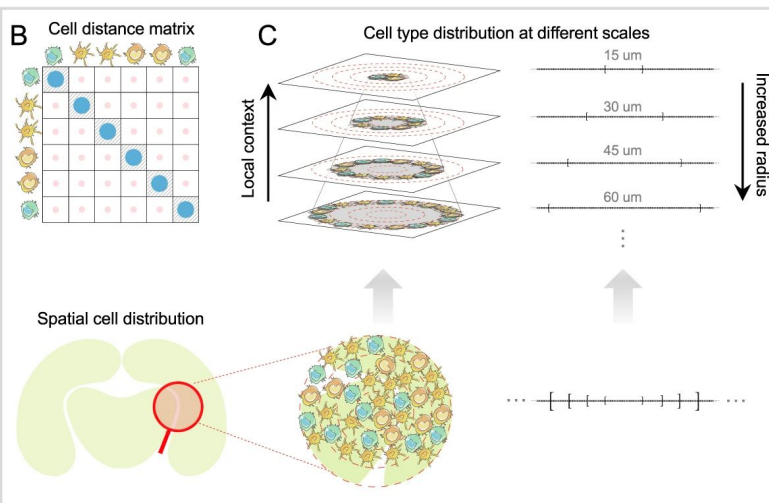
Long et al., 2023

<https://academic.oup.com/nar/article/53/7/gkaf303/8114322>

A

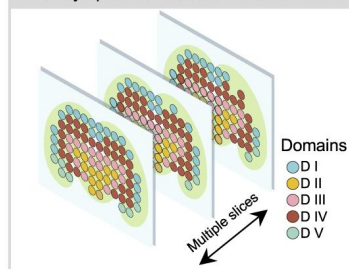


MENDER
Multi-range cEl
coNtext DEcipherER



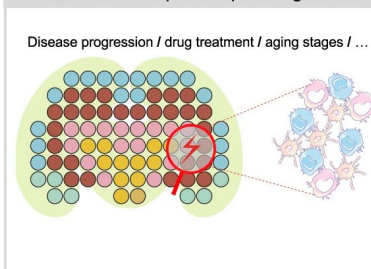
D

Identify spatial domains from multi-slice



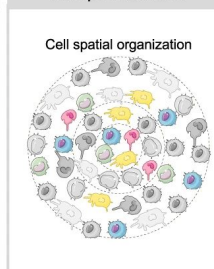
E

Detect condition-specific spatial signatures



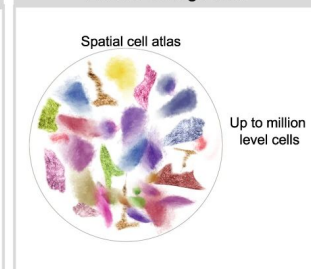
F

Interpret features



G

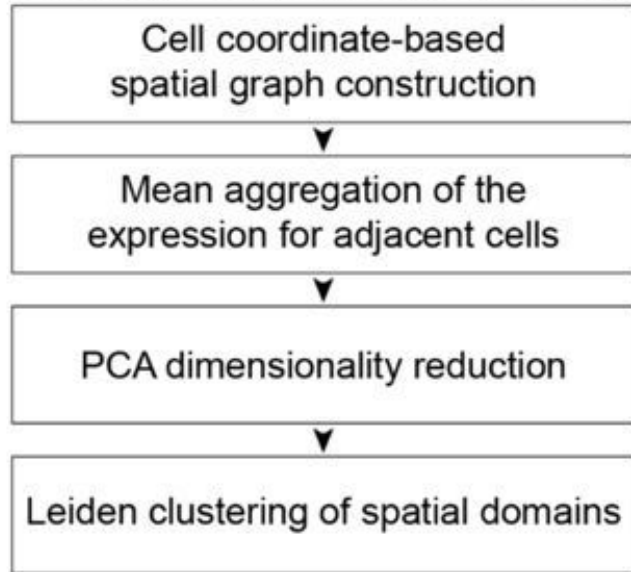
Extend to large data



MENDER

Yuan, 2024

<https://www.nature.com/articles/s41467-023-44367-9>



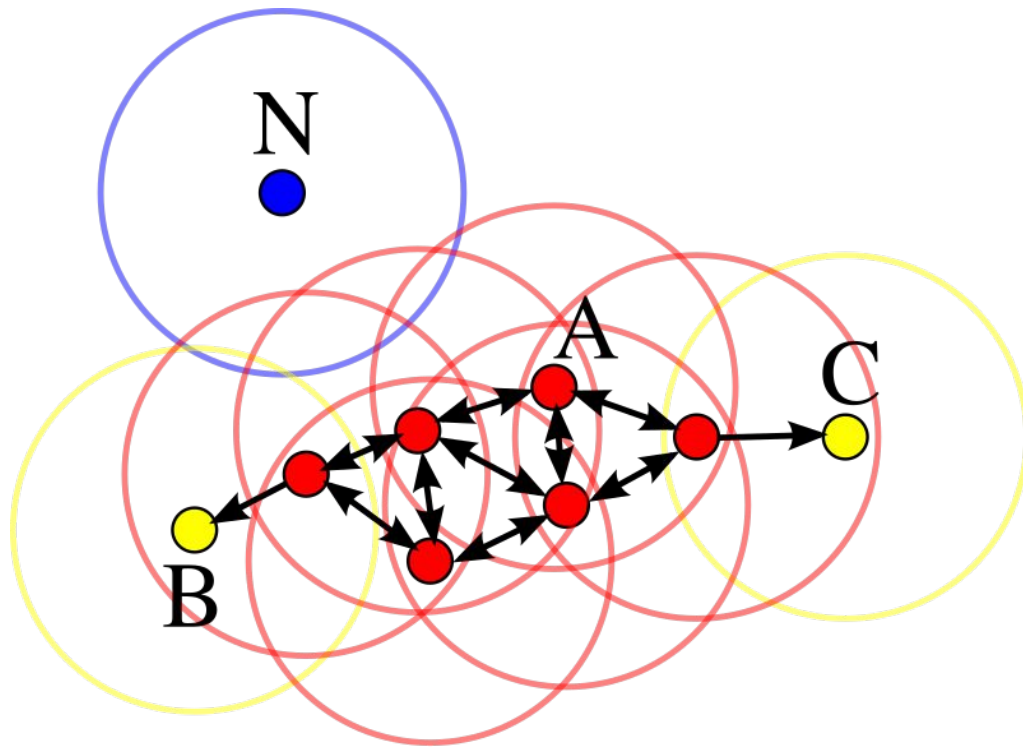
nichePCA

Schaub & Yousefi et al. 2025

<https://academic.oup.com/bioinformatics/article/41/1/btaf005/7945104>

Figure modified from

<https://www.biorxiv.org/content/10.1101/2024.12.18.629206v1.full>



DBSCAN

Ester et al., 1996

Image source: <https://en.wikipedia.org/wiki/DBSCAN>

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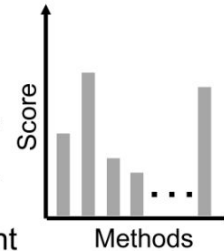
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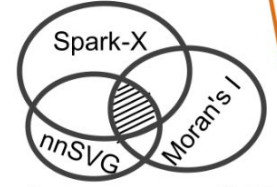
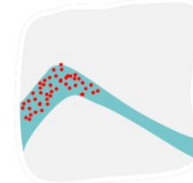
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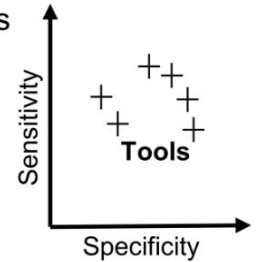
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Software:



- Provide structure for spatial data matrices and accessory data (e.g. UMAPs)
- functions for data visualizations, combining ST data with microscopy data
- standardized workflows for QC (filtering of poorly expressed genes), preprocessing, specialized techniques for ST

Function	R	Python
both well documented and big user communities	Seurat	Scanpy
format	SeuratObject	anndata
specialized workflows for spatial transcriptomics	Giotto	stLearn
Spatially variable gene identification, cell cell interaction	semLa	squidpy

Packages

Python

AnnData: <https://anndata.readthedocs.io/en/stable/>

Scanpy: <https://scanpy.readthedocs.io/en/stable/>

Squidpy: <https://squidpy.readthedocs.io/en/stable/>

Spatialdata: <https://spatialdata.scverse.org/en/stable/>

R

Seurat: <https://satijalab.org/seurat/>

Giotto: https://rubd.github.io/Giotto_site/

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Spatialdata: <https://spatialdata.scverse.org/en/stable/> [\[Not used here\]](#)

R

Seurat: <https://satijalab.org/seurat/> [\[Not used here\]](#)

Giotto: https://rubd.github.io/Giotto_site/ [\[Not used here\]](#)

Questions?